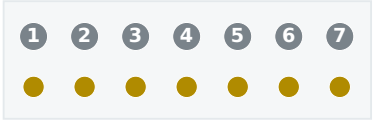
## (a) Matched team comparisons

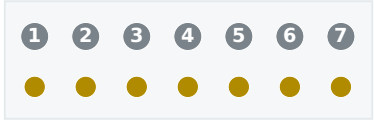
Same image, target, team size n and retained members D

**A = D ∪ X**



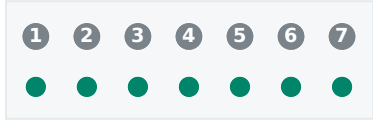

D: same seven people
X: seven C2 members

**B = D ∪ Y**




D: same seven people
Y: seven C2 members

**C = D ∪ Z**




D: same seven people
Z: seven C3 members

Within-profile replacement

Cross-profile comparison

X and Y share a profile; Z has a different profile. X, Y and Z do not overlap.

## (b) Different questions, different contrasts

**Direct member replacement**

$$\Delta_{AC} = Q_C - Q_A$$

Direction of a specified exchange

Retain D; replace X with Z

**Matched A–B–C**

$$G_S = \tfrac{1}{2}\left(|Q_C - Q_A| + |Q_C - Q_B|\right) - |Q_B - Q_A|$$

Excess change magnitude

**Direction is separate**

$$\Delta_{CB} = Q_C - Q_B$$

Positive / negative / near zero

Large magnitude need not be a gain